

NOTE(S):

LINE SOURCE.

(1.) GREEN'S FUNCTION.

$$\exp \left\{ \frac{-i k r_n}{r_n} \right\} ?$$

4

 $L \ll \lambda$; "SOMEWHAT BIGGER"

LENGTH OF ARRAY

5.

 kL is how long the line is compared to a wave length.

If $kL \gg 1$, then the line source makes cylindrical waves about the center.

~~$p \propto \frac{1}{r}$~~ $p \propto \frac{1}{(r)^{1/2}}$; CYLINDRICAL WAVES.

3dB / DOUBLING OF DISTANCE, r .

6

HEAVY FLUID.

7.

PURE MONOPOLE $\rightarrow$ WAVELENGTH IS NOT IMPORTANT.

8 BAFPLED PISTON.

NO END-RUNS BL FLUID.

10 MONOPOLES; ALL IN-PHASE.

ASSIGNMENT: DISTRIBUTION OF MONOPOLES ON SURFACE IS HARD.

11. HALF WAVELENGTH OSCILLATION.

NULLS CAN
ONLY HAPPEN

$\left\{ \begin{array}{l} x < a; \text{ PISTON RADIUS.} \end{array} \right.$

BAFFLE PISTON SOUND POWER.

W_{piston} IS THE REFERENCE; CONSTANT IS 1.

3 of 3

RADIATION EFF.

PULSATING SPHERE $\rightarrow$ NOT MONOPOLE.

$\frac{1}{r}$ $\rightarrow$ 6 dB ϕ OL DOUBLING. IN FAR FIELD (FF).